05/08/2026

CODE-C

Aakash
Medical | IIT-JEE | Foundations

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FORTNIGHTLY TEST FOR NEET-2027 (XII-PASSED)

MM: 720 **FT-OIC** **Time: 180 Mins.**

Answers

1.	(3)	37.	(2)	73.	(2)	109.	(3)	145.	(4)
2.	(3)	38.	(3)	74.	(1)	110.	(2)	146.	(2)
3.	(4)	39.	(2)	75.	(2)	111.	(3)	147.	(3)
4.	(2)	40.	(3)	76.	(4)	112.	(3)	148.	(2)
5.	(3)	41.	(2)	77.	(2)	113.	(4)	149.	(2)
6.	(3)	42.	(4)	78.	(2)	114.	(3)	150.	(4)
7.	(3)	43.	(1)	79.	(4)	115.	(1)	151.	(1)
8.	(2)	44.	(2)	80.	(3)	116.	(2)	152.	(1)
9.	(2)	45.	(2)	81.	(1)	117.	(2)	153.	(2)
10.	(2)	46.	(2)	82.	(1)	118.	(2)	154.	(4)
11.	(4)	47.	(2)	83.	(2)	119.	(4)	155.	(3)
12.	(1)	48.	(3)	84.	(4)	120.	(1)	156.	(1)
13.	(3)	49.	(3)	85.	(1)	121.	(3)	157.	(2)
14.	(2)	50.	(4)	86.	(3)	122.	(4)	158.	(3)
15.	(3)	51.	(4)	87.	(4)	123.	(1)	159.	(3)
16.	(2)	52.	(4)	88.	(2)	124.	(2)	160.	(3)
17.	(4)	53.	(3)	89.	(3)	125.	(4)	161.	(3)
18.	(1)	54.	(3)	90.	(1)	126.	(2)	162.	(3)
19.	(1)	55.	(4)	91.	(4)	127.	(1)	163.	(1)
20.	(4)	56.	(1)	92.	(3)	128.	(3)	164.	(3)
21.	(1)	57.	(1)	93.	(4)	129.	(3)	165.	(3)
22.	(3)	58.	(2)	94.	(1)	130.	(2)	166.	(2)
23.	(1)	59.	(2)	95.	(3)	131.	(4)	167.	(3)
24.	(1)	60.	(4)	96.	(4)	132.	(4)	168.	(2)
25.	(4)	61.	(2)	97.	(3)	133.	(3)	169.	(2)
26.	(3)	62.	(3)	98.	(2)	134.	(2)	170.	(2)
27.	(3)	63.	(4)	99.	(2)	135.	(3)	171.	(4)
28.	(1)	64.	(4)	100.	(1)	136.	(3)	172.	(3)
29.	(3)	65.	(4)	101.	(3)	137.	(3)	173.	(3)
30.	(4)	66.	(2)	102.	(1)	138.	(2)	174.	(2)
31.	(2)	67.	(1)	103.	(3)	139.	(1)	175.	(2)
32.	(4)	68.	(4)	104.	(4)	140.	(3)	176.	(4)
33.	(1)	69.	(1)	105.	(3)	141.	(3)	177.	(1)
34.	(4)	70.	(3)	106.	(4)	142.	(2)	178.	(4)
35.	(3)	71.	(2)	107.	(1)	143.	(4)	179.	(3)
36.	(2)	72.	(4)	108.	(2)	144.	(3)	180.	(4)

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Hints and Solutions

PHYSICS

(1) **Answer :** (3)
Solution:

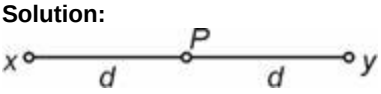
$\bar{v} = \frac{u+v}{2}$

Here $v = u + at$, so $\bar{v} = \frac{2u+at}{2}$.

(2) **Answer :** (3)
Solution:
The distance moves during first second of downward journey = $\frac{1}{2}gt^2$ which is independent of velocity of vertically upward projection.

(3) **Answer :** (4)
Solution:
At constant speed
$a = 0$
$4v^2 - 20v + 16 = 0$
$v^2 - 5v + 4 = 0$
$v^2 - 4v - v + 4 = 0$
$v(v-4) - 1(v-4) = 0$
$v = 1, 4$

(4) **Answer :** (2)
Solution:



$2 \times a \times d = v^2 - 1^2$ and $2 \times a \times d = 7^2 - v^2$
$\Rightarrow 49 - v^2 = v^2 - 1$
$\Rightarrow 2v^2 = 50$
$\Rightarrow v^2 = 25$
$\Rightarrow v = 5 \text{ m s}^{-1}$

(5) **Answer :** (3)
Solution:
$S_n = u + \frac{a}{2}(2n-1)$
$= 0 + \frac{3}{2}(2 \times 4 - 1) = 10.5 \text{ m}$

(6) **Answer :** (3)
Solution:
$a = \frac{vdv}{ds} \Rightarrow \frac{dv}{ds} = \tan(150^\circ) = \tan(90^\circ + 60^\circ)$
$= \cot 60^\circ = \frac{-1}{\sqrt{3}}$
$\Rightarrow a = -4 \times \frac{1}{\sqrt{3}} = \frac{-4}{\sqrt{3}} \text{m s}^{-2}$

(7) **Answer :** (3)
Solution:
From equations of uniformly accelerated motion,
$2 \times g \times h = (v_0)^2 \Rightarrow h = \frac{{v_0}^2}{2g}$
$\Rightarrow \frac{h}{2h} = \frac{(v_0)^2}{(v')^2} \Rightarrow (v')^2 = 2{v_0}^2$
$\Rightarrow v' = \sqrt{2}v_0$

(8) **Answer :** (2)

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Solution:
Slope of position-time (x - t) graph represents velocity.
In graph-2, the slope is increasing hence the velocity is increasing hence acceleration is positive.

(9) **Answer :** (2)
Solution:
$S_{last} = S_{three}$
$\frac{g}{2}(2t-1) = \frac{1}{2} \cdot g(3)^2$, [u = 0]
$\Rightarrow 2t - 1 = 9$
t = 5 second

(10) **Answer :** (2)
Hint:
Use equations of uniformly accelerated motion
Solution:
Here, $h = \frac{1}{2}gt^2$ [$\because u = 0$]
And $40 = \frac{1}{2}gt^2 - \frac{1}{2}g(t-2)^2$
$40 = \frac{1}{2}g[t^2 - t^2 + 4t - 4]$
$40 = 5(4t - 4)$
$40 = 20t - 20$
$\therefore t = 3$ s
Now, $h = \frac{1}{2}(10)(3)^2 = 45$ m

(11) **Answer :** (4)
Hint:
Use formula for velocity, $v = \frac{dx}{dt}$
Solution:
$v = \frac{dx}{dt} = 10t - 4$
$v = 0 = 10t - 4$
t = 0.4 second

(12) **Answer :** (1)
Hint:
Find the time until the trains collide and multiply it with the speed of the bird to find the total distance flown by the bird.
Solution:
Relative speed of trains = (72 + 54) × $\frac{5}{18}$ = 35 m/s
Time until collision = $\frac{1.75 \times 1000}{35}$ = 50 s
Total distance flown by the bird $= \left(90 \times \frac{5}{18}\right) \times 50$ = 1250 m or 1.25 km.

(13) **Answer :** (3)
Solution:
Acceleration would be zero for uniform motion.

(14) **Answer :** (2)
Solution:
From the graph it is clear that, at $t = 4$ s
$v = 0$
$0 = u - 10(4)$
$\therefore u = 40$ m/s

(15) **Answer :** (3)
Solution:
Slope of graph of displacement vs time curve gives velocity.
$\frac{V_A}{V_B} = \frac{\tan 60^\circ}{\tan 30^\circ} = \frac{\sqrt{3}}{\frac{1}{\sqrt{3}}} = \frac{3}{1}$

(16) **Answer :** (2)
Solution:
$X = t^3 + t^2$
$v = \frac{dx}{dt} = 3t^2 + 2t$

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$v = 3(2)^2 + 2(2)|_{t=2s}$

$v = 12 + 4 = 16\,\text{ms}^{-1}$

$V_{\text{avg}} = \frac{X(2) - X(0)}{2-0}$

$= \frac{8+4-0}{2} = 6\,\text{ms}^{-1}$

$\frac{v}{v_{\text{avg}}} = \frac{16}{6} = \frac{8}{3}$

(17) Answer : (4)
Solution:

$T = \frac{7+3}{2} = 5\text{ s}$

$V = U - gT$

$O = U - 10 \times 5$

$U = 50\text{ m s}^{-1}$

(18) Answer : (1)
Hint:

Least count = $\frac{\text{Pitch}}{\text{Number of divisions on circular scale}}$

Solution:

Pitch = $\frac{2\text{ mm}}{2\text{ rotations}} = 1\text{ mm}$

$\text{L} \cdot \text{C} \cdot = \frac{1\text{ mm}}{100} = 0.01\text{ mm}$

(19) Answer : (1)
Solution:
When two measured values are multiplied or divided, there should be as many significant figure retained in the final results as are there in the original number with least significant figures.
414.4 × 94 = 38953.6
= 3.9×10^4

(20) Answer : (4)
Solution:
[Q] = [It] = [AT]

(21) Answer : (1)
Solution:
Velocity is a physical quantity which has unit as well as dimension.

(22) Answer : (3)
Solution:

$\%T = \frac{1}{2}(\%l + \%g)$

$\%T = \frac{1}{2}(2+4)$

$\%T$ = 3%

(23) Answer : (1)
Solution:

$$n_2 = n_1 \left(\frac{M_1}{M_2}\right)\left(\frac{L_1}{L_2}\right)\left(\frac{T_1}{T_2}\right)^{-2}$$

$$= 10\left(\frac{1\text{ kg}}{10\text{ kg}}\right)\left(\frac{1\text{ m}}{10\text{ m}}\right)\left(\frac{1\text{ s}}{10\text{ s}}\right)^{-2}$$

$$n_2 = 10$$

(24) Answer : (1)
Solution:
Number of insignificant zeroes in 0.0023400 are 3.

(25) Answer : (4)
Solution:
least count is the smallest measurement an instrument can take.

(26) Answer : (3)

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Solution:

$\frac{\Delta Q}{Q} \times 100 = \frac{\Delta a}{a} \times 100 + \frac{1}{2}\frac{\Delta b}{b} \times 100 + \frac{3}{2}\frac{\Delta c}{c} \times 100 + 4\frac{\Delta d}{d} \times 100$

$= \left(2 + \frac{3}{2} + \frac{3}{2} + 2\right)\%$

= 7%

(27) Answer : (3)
Solution:
Among the given quantities, velocity is not treated as fundamental unit.

(28) Answer : (1)
Solution:

$F = \propto \eta^a r^b v^c$

$\Rightarrow \left[MLT^{-2}\right] = \left[ML^{-1}T^{-1}\right]^a [L]^b \left[LT^{-1}\right]^c$

$\left[MLT^{-2}\right] = [M]^a [L]^{-a+b+c} [T]^{-a-c}$

$\Rightarrow a = 1,$

$-a-c = -2 \Rightarrow a + c = 2 \Rightarrow c = 1$ as $a = 1$

$b + c - a = 1$

$b + c = 2$, as $c = 1$ $\Rightarrow b = 1$

$\Rightarrow F \propto \eta r v$

(29) Answer : (3)
Solution:

$[\beta xt] = [M^0 L^0 T^0]$

$[\beta] = \frac{1}{(xt)} = \left[L^{-1}T^{-1}\right] = \left[M^0 L^{-1} T^{-1}\right]$

(30) Answer : (4)
Solution:

Velocity gradient $\Rightarrow \frac{\left[LT^{-1}\right]}{[L]} = \left[M^0 L^0 T^{-1}\right]$

Pressure gradient, $\frac{\frac{N}{m^2}}{m} = \frac{N}{m^3} = \frac{\left[MLT^{-2}\right]}{M^0 L^3 T^0} = \left[ML^{-2}T^{-2}\right]$

(31) Answer : (2)
Solution:

$F = \frac{mv^2}{r}$ is dimensionally correct

(32) Answer : (4)
Solution:

$1\text{g cm}^{-3} = \frac{1 \times 10^{-3}\text{ kg}}{10^{-6} m^3} = 10^3\text{ kg}/m^3$

(33) Answer : (1)
Solution:

$5 \times 1\ m^2 = \beta \times 100\ m^2$

$\beta = 0.05$

(34) Answer : (4)
Solution:
Unit of plane angle is radian while that of solid angle is steradian.

(35) Answer : (3)
Solution:
In multiplication of two physical quantities, there should be as many significant figures retained in final result, as there in the original number with the least significant figures.

(36) Answer : (2)
Solution:
- Area under velocity-time graph → Gives displacement
- Area under acceleration-time graph → Gives change in velocity
- Slope of velocity-time graph → Gives acceleration
- Slope of position-time graph → Gives velocity

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(37) Answer : (2)
Solution:
V_{AB} = 40 ms^{-1}
V_{BG} = 25 ms^{-1} [V_{BG} $\rightarrow$Velocity of train *B* w.r.t. to ground]
V_{AG} = 15 ms^{-1} [V_{AG} $\rightarrow$Velocity of train *A* w.r.t. to ground]
a_{AB} = 0

(38) Answer : (3)
Solution:
slope of position time graph gives velocity.

(39) Answer : (2)
Solution:
Normally dimensionless quantities have no units, but angle is exception.

(40) Answer : (3)
Solution:

• Any body moving under constant acceleration may change its direction of motion.
• Body may be at instantaneous rest while having non-zero acceleration at that point.
• $\left|\overrightarrow{v}_{\text{inst}}\right| = v_{\text{inst}}$

(41) Answer : (2)
Solution:
If velocity of one particle w.r.t other particle is along the line joining them, then only the two particles can meet.

(42) Answer : (4)
Solution:
$\int_0^{\pi/4} \cos 2x = \frac{1}{2}[\sin 2x]_0^{\pi/4}$
= $\frac{1}{2}\left[\sin\frac{\pi}{2} - \sin 0\right] = \frac{1}{2}$

(43) Answer : (1)
Solution:
At any time *t*, particle cannot have two positions.

(44) Answer : (2)
Solution:
Separation first decreases then increases.

(45) Answer : (2)
Solution:
Displacement = – 3 – 1 = – 4 m
Total time = 6
Total displacement magnitude = 4 m
Average velocity $= \frac{4}{6} = \frac{2}{3}\ \text{m/s}$

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CHEMISTRY

(46) Answer : (2)
Solution:
1.1 has only two significant figures

(47) Answer : (2)
Solution:
Number of electrons in one mole SO_4^{2-} ion = 50 moles.
9.6 g of SO_4^{2-} ions = $\frac{9.6}{96} = 0.1$ mole
Number of electrons = 0.1 × 50 × N_A = 5 N_A

(48) Answer : (3)

Solution:
Precision refers to the closeness of various measurements for the same quantity.
Accuracy refers to agreement of a particular value to the true value of the result.

(49) **Answer :** (3)
Solution:

Compound	Molecular formula	Empirical formula
Benzene	C_6H_6	CH
Ethyne	C_2H_2	CH
Cyclohexane	C_6H_{12}	CH_2
Propyne	C_3H_4	C_3H_4

(50) **Answer :** (4)
Solution:
Equal volumes of NH_3 and CO_2 have equal moles at STP and equal number of molecules but their masses remain different.

(51) **Answer :** (4)
Hint:
One molecule of Biomolecule should contain one atom of Zn.
Solution:

$\%\,of\,Zn = \frac{Number\,of\,Zn\,atom \times 65.38}{wt.\,of\,biomolecule} \times 100$

$0.75 = \frac{1 \times 65.38}{MW} \times 100$

∴ Minimum molar mass of biomolecule

$= \frac{65.38 \times 100}{0.75}$

= 8717.33 g/mol

(52) **Answer :** (4)
Solution:

yocto – 10^{-24}

zepto – 10^{-21}

femto – 10^{-15}

pico – 10^{-12}

(53) **Answer :** (3)
Solution:
Let the percentage abundances of two isotopes are x and (100 – x) respectively.

$12.2 = \frac{\left(x \times 12\right) + \left(100 - x\right) \times 13}{100}$

$\Rightarrow$ 1220 = 12x + 1300 – 13x

$\Rightarrow$ x = 80%

∴ Percentage abundance of 2nd isotope
= 100 – 80 = 20%

(54) **Answer :** (3)
Solution:

1 mole of atomic H = 6.022×10^{23} hydrogen atoms.

(55) **Answer :** (4)
Solution:
According to Dalton's atomic theory, chemical reactions involve reorganisation of atoms.
All the atoms of a given element have identical properties including identical mass.

(56) **Answer :** (1)
Solution:
Formula mass is applicable for ionic solid which does not exist as a single entity.

(57) **Answer :** (1)
Solution:

$\text{Mass of } CH_4 = \frac{16 \times 3.01 \times 10^{23}}{6.02 \times 10^{23}} = 8\text{g}$

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(58) **Answer :** (2)
Solution:

$$\underset{2\text{ Volume}}{2H_2(g)} + \underset{1\text{ Volume}}{O_2(g)} \rightarrow 2H_2O$$

(59) **Answer :** (2)
Solution:
Let atomic weight of A and B are 'a' and 'b' respectively.

$$0.1 = \frac{6}{\text{Atomic weight of AB}}$$

$$0.5 = \frac{40}{\text{Atomic weight of}_2\text{AB}}$$

a + b = 60 ...(i)
a + 2b = 80 ...(ii)
b = 20u
a = 40u

(60) **Answer :** (4)
Hint:

Mole of molecule/atom = $\frac{\text{Mass of molecule / atom}}{\text{Molar mass of molecule / atom}}$

Solution:
Mole of He = $\frac{2x}{4}$
Mole of N_2 = $\frac{x}{28}$

$$\frac{\text{Mole of He}}{\text{Mole of } N_2} = \frac{\frac{2x}{4}}{\frac{x}{28}} = \frac{2x}{4} \times \frac{28}{x} = 14:1$$

(61) **Answer :** (2)
Hint:
One molecule of CO_2 contains 22 electrons.
Solution:
Number of atoms = Number of molecules × 3

$$= \frac{4.4\times10^{26}}{22\times N_A} = \frac{x}{3\times N_A}$$

$= 6 \times 10^{25}$

(62) **Answer :** (3)
Solution:

x mole of Na_2SO_4 = mole of H_2O

$$\text{or} \quad x \times \frac{44.1}{142} = \frac{55.9}{18}$$

$$\text{or} \quad x = \frac{142\times55.9}{44.1\times18}$$

$$\cong 10$$

(63) **Answer :** (4)
Solution:
Percentage of oxygen in
a - $CaCO_3 = \frac{1}{100} \times 48 \times 100 = 48\%$
b - $NaOH = \frac{1}{40} \times 16 \times 100 = 40\%$
c - $CaO = \frac{1}{56} \times 16 \times 100 = 28.5\%$
d - $C_6H_{12}O_6 = \frac{1}{180} \times 96 \times 100 = 53.3\%$

(64) **Answer :** (4)
Solution:
(i) Atoms of Ca in $80\text{ amu} = \frac{80}{40} = 2$

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(ii) At STP, $22400\ \text{mL}\ \ CO_2 \equiv 3N_A$ atoms

$\Rightarrow 2240\ \text{mL}\ = \frac{2240\times 3N_A}{22400} = 0.3N_A$ atoms.

(iii) $10 = \frac{\text{Wt. of solute}}{\text{Wt. of solution}} \times 10^6$

$\therefore \frac{\text{Wt. of solute}}{\text{Wt. of solution}} = 10 \times 10^{-6}$

% by mass $= 10 \times 10^{-6} \times 100 = 10^{-3}$

(65) Answer : (4)
Hint:

Number of moles = $\frac{Given\ mass}{molar\ mass}$

Solution:

Number of atoms in 1 g $C = \frac{1}{12}\ N_A$

Number of atoms in $1\ g\ O_2 = 2 \times \frac{1}{32} N_A = \frac{1}{16}\ N_A$

Number of atoms in $1\ g\ N_2 = 2 \times \frac{1}{28}\ N_A = \frac{1}{14}\ N_A$

Number of atoms in $1\ g\ H_2 = 2 \times \frac{1}{2}\ N_A = \ N_A$

(66) Answer : (2)
Solution:

$CaCO_3(s) + 2HCl\ (aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

Moles of HCl = $\frac{100 \times 0.5}{1000} = 0.05$ mol

As per reaction stoichiometry:

2 mol of HCl $\equiv$ 1 mol of pure $CaCO_3$.

$\therefore$ 0.05 of HCl $\equiv \frac{1}{2} \times 0.05 = \frac{1}{40}$mol

Mass of pure $CaCO_3$ needed

= $\frac{1}{40} \times 100 = 2.5$ g

Mass of sample (impure needed)

= $\frac{100}{80} \times 2.5 = 3.125$ g

(67) Answer : (1)
Solution:

$P_4 + 5O \rightarrow P_4O_{10}$

31g

$\frac{1}{4}$ mol P_4 gives $\frac{1}{4}$ mol P_4O_{10} for 100% yield or $\frac{1}{4} \times \frac{60}{100}$ mol P_4O_{10} when 60% yield

Mol of $P_4O_{10} = \frac{1}{4} \times \frac{60}{100}$ mol

Mass of P_4O_{10} = n × M_0 = $\frac{1}{4} \times \frac{60}{100} \times 284 = 42.6\ g$

(because MW (P_4O_{10}) = (31 × 4) + (16 ×10) = 284)

(68) Answer : (4)
Solution:

1 mol N_2 $\equiv$ 2 mol NO $\equiv$ 2 mol NO_2

$\therefore$ 0.5 mol N_2 $\equiv$ 1 mol NO $\equiv$ 1 mol NO_2

Mass of NO_2 = 1 × 46 = 46 g

(69) Answer : (1)
Solution:

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

1 mol = 2 mol

(16 g) (64 g)

$\therefore$ 8 g of CH_4 will react with

2 × 0.5 $\Rightarrow$ 1 mol of O_2 = 22.4L

So, no limiting reagent in the reaction

Mass of CO_2 = 22 g

On reducing the masses of both the reactants to half,

Mass of CH_4 = $\frac{8}{2} = 4$ g

Mass of 22.4 L of O_2 = $\frac{22.4}{22.4}$L = 1 mol

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= 32 g O_2

$\therefore$ On reducing mass of O_2 to half = $\frac{32}{2} = 16\,g$

CH_4 + $2O_2$ $\rightarrow$ CO_2 + $2H_2O$

(16 g) + (64 g) (44 g)

$\therefore$ 8 g CH_4 reacts with 32 g O_2 to form 22 g CO_2

(70) **Answer :** (3)

Solution:

	$CO_2(g)$	+	$C(s)$	$\longrightarrow$	$2CO(g)$
Initial	1L		0		0
Final volume	$(1-x)$				2x

Final volume = 1 – x + 2x = 1.4 L

$\therefore$ $\boxed{x = 0.4L}$

$\therefore$ Volume of CO = 2x = 2 × 0.4 = 0.8 L

Volume of CO_2 = (1 – x) = 1 – 0.4 = 0.6 L

(71) **Answer :** (2)

Hint:

$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$

Solution:

$\because$ 2 mol $NaHCO_3$ gives 1 mol $CO_2(g)$

$\therefore$ 0.1 mol $NaHCO_3$ will give 0.05 mol CO_2

= 0.05 × 22.4 L CO_2 at STP

= 1.12 L

(72) **Answer :** (4)

Solution:

	H_2	+	Cl_2	$\rightarrow$	2HCl
	1 Vol		1 Vol		2 Vol
Given	10L		20L		
	$10L\,H_2$	=	$10L\,Cl_2$	=	$20L\,HCl$
After reaction	H_2 gets consumed		10 L Cl_2 remains in reaction mixture		20 L HCl is produced

(73) **Answer :** (2)

Hint:

$\underset{(s)}{Mg} + \underset{(aq.)}{H_2SO_4} \rightarrow \underset{(aq.)}{MgSO_4} + \underset{(g)}{H_2}$

Solution:

Mole of Mg = mole of H_2

$\frac{wt.}{24} = \frac{112}{22400}$

wt. of Mg = 0.12 g

(74) **Answer :** (1)

Solution:

$C_6H_{12}O_6 + 6O_2 \rightarrow 6\,CO_2 + 6H_2O$

Mole of CO_2 = $\frac{44}{44} = 1$

Mole of glucose required = $\frac{1}{6}$

Mass of glucose required = $\frac{1}{6} \times 180 = 30\,g$

(75) **Answer :** (2)

Solution:

As per the law of multiple proportions, if two elements combine to form more than one compound the masses of one element that combine with a fixed mass of the other element, are in the ratio of small whole numbers.

(76) Answer : (4)
Hint:
$x_2 = \frac{n_2}{n_2 + n_1}$
Solution:
$x_1 = 0.85$, $x_2 = 0.15$,
Let number of moles of $H_2SO_4 = n_{H2SO4}$
$x_{H_2SO_4} = \frac{n_{H_2SO_4}}{n_{H_2O} + n_{H_2SO_4}}$
$0.15 = \frac{n_{H_2SO_4}}{\frac{2\times10^3}{18} + n_{H_2SO_4}}$
x = 19.6

(77) Answer : (2)
Solution:
Mass of solvent in solution = 1000 g
Mole of solute = 1
Mass of solute = 1 × 180 = 180 g
Mass percentage of solute $= \frac{\text{mass of solute}}{\text{mass of solution}} \times 100$
$= \frac{180}{1180} \times 100 \approx 15.25\%$

(78) Answer : (2)
Solution:
$x = \frac{4.5}{4.5+55.5} = \frac{4.5}{60} = \frac{3}{40}$

(79) Answer : (4)
Hint:
Number of mole(s) $= \frac{\text{Number of molecules}}{N_A}$
Solution:
Mole of sugar $= \frac{3.01\times10^{23}}{6.02\times10^{23}} = 0.5$
Mole fraction of sugar $= \frac{0.5}{0.5+24.5} = 0.02$

(80) Answer : (3)
Hint:
$\chi_{CH_3OH} = \frac{n_{CH_3OH}}{n_{CH_3OH} + n_{C_2H_5OH}}$
Solution:
Let mass of CH_3OH and C_2H_5OH taken is w gram
$X_{CH_3OH} = \frac{\frac{w}{32}}{\frac{w}{32} + \frac{w}{46}} = \frac{\frac{w}{32}}{\frac{23w+16w}{736}}$
$= \frac{w}{32} \times \frac{736}{39w} = \frac{23}{39}$

(81) Answer : (1)
Solution:
In 1 kg of 0.1 molal glucose solution, mass of glucose = 180 g
∴ Mass of solution = 1000 + 180 = 1180
⇒ 1180 g of glucose solution ≡ 180 g glucose
∴ 2000 g of glucose solution $= \frac{180}{1180} \times 2000 \approx 305g$

(82) Answer : (1)
Hint:
$M_1V_1 + M_2V_2 = M_3V_3$.
Solution:
$M_3 = \frac{M_1V_1 + M_2V_2}{V_3}$
$= \frac{0.1\times2 + 0.2\times1}{2+1} = 0.133\ M$

(83) Answer : (2)

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Solution:

Moles of urea = $400 \times 2.5 \times 10^{-3}$

Number of urea molecules = $400 \times 2.5 \times 10^{-3} \times 6.02 \times 10^{23}$

$= 6.02 \times 10^{23}$

(84) **Answer :** (4)

Solution:

$$BaCl_2(aq) + H_2SO_4(aq) \rightarrow BaSO_4 \downarrow + 2HCl(aq)$$

Mole t = 0	$\frac{20.8}{208} = 0.1$	$\frac{14.7}{98} = 0.15$	0	0
Mole after reaction	0	0.05	0.1	0.2

$\therefore \ [SO_4^{2-}] = \frac{0.05}{0.2} = 0.25$ M

(85) **Answer :** (1)

Hint:

Molarity = $\frac{no.\ of\ moles\ of\ solute}{volume\ of\ solution\ in\ L}$

Solution:

Number. of moles of methanol required = M × V

= 0.3 × 3 = 0.9

Mass of methanol required = moles × molar mass

= 0.9 × 32 = 28.8 g

Volume of methanol = $\frac{mass}{density} = \frac{28.8}{0.793} = 36.32\ mL$

(86) **Answer :** (3)

Hint:

$Molality = \frac{Moles\ of\ solute \times 1000}{Weight\ of\ solvent(g)}$

Solution:

$Molality = \frac{0.1 \times 1000}{0.9 \times 18} = \frac{1000}{9 \times 18} = 6.17\ m$

Molality of H^+ = 6.173 × 2 = 12.346 ; 12.34 m

Molality of SO_4^{2-} = 6.17 m

(87) **Answer :** (4)

Solution:

% by mass = $\frac{10}{10^6} \times 100 = 10^{-3}$.

(88) **Answer :** (2)

Solution:

$m = \frac{x_{solute}}{1 - x_{solute}} \times \frac{1000}{M_{solvent}}$

(89) **Answer :** (3)

Solution:

$3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$

0.3 mole of $BaCl_2$ combines with 0.2 mole of Na_3PO_4. Therefore, $BaCl_2$ is limiting reactant.

0.3 mole of $BaCl_2$ will produce 0.1 mole of $Ba_3(PO_4)_2$ and 0.6 mole of NaCl.

(90) **Answer :** (1)

Solution:

$$2KClO_3 \longrightarrow 2KCl + 3O_2$$

2 moles 3 moles

2 mol of $KClO_3$ gives = 3 mol O_2

1 mol of $KClO_3$ gives = $\frac{3}{2}$ mol O_2

For Al burning

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$$2\ Al\ +\ \frac{3}{2}\ O_2\ \rightarrow\ \underset{1\ mole}{3O_2}$$

As $\frac{3}{2}$ mole of O_2 gives 1 mole Al_2O_3

$\therefore$ 1 mole Al_2O_3 formed.

BOTANY

(91) **Answer :** (4)
Solution:
Prokaryotic cells lack nuclear membrane.

(92) **Answer :** (3)
Solution:
Rudolf Virchow first explained that cells divided and new cells are formed from pre-existing cells.

(93) **Answer :** (4)
Solution:
The endomembrane system include endoplasmic reticulum (ER), golgi complex, lysosomes and vacuoles.

(94) **Answer :** (1)
Solution:
Tonoplast, in which is the membrane surrounding the vacuole, actively transports ions into the vacuole, against their concentration gradient. Active transport by the tonoplast results in a higher concentration of ions in the vacuole compared to the cytoplasm.

(95) **Answer :** (3)
Solution:
The cell wall of a young plant cell, the primary wall is capable of growth.The middle lamella is a layer mainly of calcium pectate which holds or glues the different neighbouring cells together.

(96) **Answer :** (4)
Solution:
Mycoplasmas, the smallest cells, are only 0.3 µm in length while bacteria could be 3 to 5 µm. The largest isolated single cell is the egg of an ostrich.

(97) **Answer :** (3)
Solution:
The cell wall and middle lamellae may be traversed by plasmodesmata which connect the cytoplasm of neighbouring cells. On the sides of centromere there are disc shaped structures called
kinetochores.

(98) **Answer :** (2)
Solution:
The shape of nerve cells is branched and long.

(99) **Answer :** (2)
Hint:
Secondary constriction is a non-staining region of SAT chromosome.
Solution:
Satellite is a region beyond secondary constriction in SAT chromosome.

(100) **Answer :** (1)
Solution:
Several ribosomes may attach to a single mRNA and form a chain called polyribosomes or polysome.

(101) **Answer :** (3)
Solution:
Higher plants cell lack centrioles but it is found in animals cells.

(102) **Answer :** (1)
Solution:
All prokaryotes have a cell wall surrounding the cell membrane except in mycoplasma.

(103) **Answer :** (3)
Solution:

The major function of contractile vacuole is osmoregulation, *i.e.,* removal of excess of water.

(104) Answer : (4)
Solution:
Rod shaped - Bacillus, spherical-Coccus, spiral-Spirillum, comma shaped - Vibrium.

(105) Answer : (3)
Solution:
Mesosomes are infoldings of plasma membrane. Mesosomes involved in cell wall formation, DNA Replication, Respiration.

(106) Answer : (4)
Solution:
Processes like secretion and endocytosis are functions associated with plasma membrane.

(107) Answer : (1)
Solution:
Nucleolus and cell membrane are usually found in both plant and animal cells.

(108) Answer : (2)
Solution:
A specialised differentiated form of cell membrane called mesosome is the characteristic of prokaryotes. They are essentially infoldings of cell membrane.

(109) Answer : (3)
Solution:
Centrioles and ribosomes lack membrane

(110) Answer : (2)
Solution:
Lysosomal enzymes digest nucleic acids too

(111) Answer : (3)
Solution:
Cytoskeleton is present in eukaryotes.

(112) Answer : (3)
Hint:
Prokaryotic flagella is made up of flagellin protein.
Solution:
A typical bacterial flagellum has three components: hook, filament and basal body.

(113) Answer : (4)
Solution:
Eukaryotes have both 70S and 80S ribosomes

(114) Answer : (3)
Solution:
Antonie Von Leeuwenhoek first saw and described a live cell.

(115) Answer : (1)
Solution:
Organelles that constitute endomembrane system are; Golgi complex, ER, lysosome and vacuoles. Ribosomal RNA synthesis occurs in nucleolus which is not the part of endomembrane system.

(116) Answer : (2)
Solution:
(i) Ribosomes are made up of RNA and proteins
(ii) Primary wall is capable of growth
(iii) Endomembrane system shows coordinated function

(117) Answer : (2)
Solution:
In some prokaryotes like cyanobacteria, there are other membranous extensions into the cytoplasm called chromatophores which contain pigments.

(118) Answer : (2)
Solution:
The axoneme of a eukaryotic flagellum typically has a "9+2" arrangement, consisting of nine peripheral microtubule doublets and two central microtubules. Each of the nine peripheral doublets is connected to its neighboring doublet by interdoublet linkers. Therefore, the number of interdoublet linkers is equal to the number of connections between these doublets, which is 9.

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(119) Answer : (4)
Solution:
G. Palade - Ribosome

(120) Answer : (1)
Solution:

Structure A (nucleus) contains chromatin which contain DNA, RNA, histones and non-histone protein.
Structure B (Nuclear pore) formed due to fusion of two membranes of nuclear envelope.
Structure C (RER), site for protein synthesis and secretion.
Structure D(SER) site for lipid synthesis.

(121) Answer : (3)
Solution:
Centrosome - An organelle containing two cylindrical structures which are surrounded by amorphous pericentriolar material.
Centrioles - Made up of nine evenly spaced peripheral fibrils of tubulin, and forms the basal body of flagella.
Chromatin - Is highly extended and elaborate nucleoprotein fibres.

(122) Answer : (4)
Solution:
Chemical reactions occurring in the cytoplasm keep the cell in living state.
Centriole is a membraneless structure which is absent in higher plants.
Theodore Schwann reported that animal cells have a thin outer layer.

(123) Answer : (1)
Solution:
i) There are 9 radial spokes each in cilia/flagella and centrioles.
ii) Cilia are made up of microtubules coated by the plasma membrane, whereas, centrioles are not membrane bound.

(124) Answer : (2)
Hint:
Telocentric chromosomes have centromere located at the terminal end of chromosome.
Solution:
Metacentric chromosome has a centromere located at the middle, resulting in two equal arms. Sub-metacentric chromosome has centromere slightly away from the middle, resulting in one shorter and one longer arm. Acrocentric chromosome has centromere located close to one end, forming one short and one long arm.

(125) Answer : (4)
Solution:
The chloroplasts contain chlorophyll and carotenoid pigments which are responsible for trapping light energy essential for photosynthesis

(126) Answer : (2)
Solution:
Oil storage : Elaioplast
Oxysomes : Mitochondria
Protein storage : Aleuroplast
Light reaction: Chloroplast

(127) Answer : (1)
Solution:
Cisternae in Golgi apparatus are interconnected.

(128) Answer : (3)
Solution:
Capsule is a thick and tough covering found in the cell envelope of bacteria. It is made up of polysaccharides.

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(129) Answer : (3)
Solution:
Chromosome is divided into one shorter (p arm, petit = small) and one longer (q arm). These arms can be seen in sub-metacentric and acrocentric chromosomes.

(130) Answer : (2)
Solution:
Fungal cell wall is generally composed of chitin. Plasmodesmata is found in plant cells.

(131) Answer : (4)
Solution:
The lipids are arranged within the membrane with the polar head towards the outer sides and the hydrophobic tails towards the inner part.This ensures that the nonpolar tail of saturated hydrocarbons is protected from the aqueous environment

(132) Answer : (4)
Solution:
In plants, the middle lamella is a layer mainly composed of calcium pectate which holds or glues the different neighbouring cells together.

(133) Answer : (3)
Solution:
Fimbriae present in bacteria are small bristle-like fibres sprouting out of the cell and help in attachment.

(134) Answer : (2)
Solution:
In human beings, the membrane of the erythrocyte has approximately 52% proteins and 40% lipids.

(135) Answer : (3)
Solution:
Both mesosome and mitochondria play important role in respiration.

ZOOLOGY

(136) Answer : (3)
Solution:
Tight junctions help to stop substances from leaking across a tissue. Adhering junctions perform cementing to keep neighbouring cells together.

(137) Answer : (3)
Solution:
Generally, neuroglia are smaller than neurons, and they are 5 to 50 times more numerous. In contrast to neurons, glia do not generate or propagate action potentials, and they can multiply and divide in the mature nervous system.

(138) Answer : (2)
Solution:
Smooth muscle fibres are found in visceral organs. The smooth muscle fibres taper at both ends (fusiform) and do not show striations.

(139) Answer : (1)
Solution:
Brush bordered cuboidal epithelium is present in PCT.

(140) Answer : (3)
Solution:
Columnar or cuboidal epithelial cells that bear cilia on their free surface are called ciliated epithelium. The cilia remain in rhythmic motion and create a current to transport material which comes in contact with them. In respiratory tract, it helps in expelling mucus and trapped particles towards outside.

(141) Answer : (3)
Solution:
Bones and cartilages are specialised connective tissues. Calcium salts and collagen fibres are present in bone matrix and give bone its strength.

(142) Answer : (2)
Hint:
Multicellular organism with tissue level of organisation
Solution:

Hydra is made of different types of cells and the number of cells in each type can be in thousands.

(143) Answer : (4)
Hint:
These muscle fibres taper at both ends.
Solution:
Smooth or visceral muscle fibres are fusiform in shape while skeletal muscle fibres are cylindrical.

(144) Answer : (3)
Solution:
Neural tissue exerts the greatest control over the body's responsiveness to changing conditions.

(145) Answer : (4)
Hint:
Structure that connects bone to bone.
Solution:
Ligaments and tendons are the examples of dense regular connective tissues. Areolar and adipose tissues are examples of loose connective tissue. Dense irregular connective tissue has fibroblasts and many fibres (mostly collagen) that are oriented differently. This tissue is present in the skin.

(146) Answer : (2)
Hint:
Cardiac muscle fibres.
Solution:
Intercalated discs are communication junctions that allow the cardiac cells to contract as a unit.

(147) Answer : (3)
Solution:
Alveoli of lungs are lined by simple squamous epithelium.

(148) Answer : (2)
Solution:
Fibroblasts secrete collagen fibres.

(149) Answer : (2)
Solution:
Tendons and ligaments are examples of dense regular connective tissue.

(150) Answer : (4)
Solution:
Lymph is formed from the filtration of blood. Blood is a fluid connective tissue containing plasma, red blood cells (RBCs), white blood cells (WBCs) and platelets. Blood and lymph lack connective tissue fibers. (collagen, elastic and reticular).

(151) Answer : (1)
Solution:
Germinal epithelium of ovary and seminiferous tubule is made of cuboidal epithelial cells.

(152) Answer : (1)
Hint:
Chondrocytes are present in this tissue.
Solution:
Cartilage is present in the tip of nose. Tendon and ligament are dense regular connective tissues.
Bone and cartilage are specialized connective tissues.

(153) Answer : (2)
Solution:
In neural tissues, neurons coordinate between different organs through nerve impulses and it reacts or adjusts the body with respect to change in the environment.

(154) Answer : (4)
Solution:
In skeletal muscles (*e.g.*, Biceps) many nuclei occur at irregular intervals and are peripheral in position. These fibres are multinucleated responsible for structural syncytium. *e.g*. thigh muscles.

(155) Answer : (3)
Hint:
A type of this tissue links and supports different tissues.
Solution:
Connective tissue is most abundant tissue in complex organisms.

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(156) Answer : (1)
Solution:
Goblet cells of the alimentary canal secrete mucus.

(157) Answer : (2)
Hint:
These cells are tall and have basal nuclei.
Solution:
Cilia facilitate movement of mucus in a specific direction.

(158) Answer : (3)
Solution:
Sweat, sebum and saliva are secretions of exocrine glands *i.e.*, they have ducts.

(159) Answer : (3)
Solution:
Areolar tissue acts as a filling tissue in body and it is also used as packing material between organs, whereas adipose tissue primarily acts as thermal insulator and fat reservoir. Both areolar tissue and adipose tissue possess mast cells and adipocytes.

(160) Answer : (3)
Solution:
Neural tissue shows excitability. Cells of connective tissues are not excitable.

(161) Answer : (3)
Solution:
Vinblastin is used to treat a number of cancers.

(162) Answer : (3)
Solution:
In a polysaccharide chain (like glycogen), the right end is called the reducing end and the left end is called the non-reducing end. The reducing end of a sugar has free aldehydic group, when its ring structure opens.

(163) Answer : (1)
Solution:
Glycogen and starch are homopolymers of glucose.

(164) Answer : (3)
Solution:
The starch-I_2 complex is blue in colour.

(165) Answer : (3)
Solution:
Lipids are obtained in acid-insoluble fraction / retentate fraction.

(166) Answer : (2)
Solution:
• For analysis of organic compounds in living organisms, a living tissue is grinded in trichloroacetic acid (Cl_3CCOOH) using a mortar and a pestle.

(167) Answer : (3)
Solution:
Organic compounds get oxidised to CO_2 and H_2O when tissue is burnt.

(168) Answer : (2)
Solution:

Element	% weight of human body
Hydrogen	9.5
Carbon	18.5
Oxygen	65.0
Calcium	1.5

(169) Answer : (2)
Solution:
Cellulose is used for making paper.

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(170) Answer : (2)

Solution:

Cellulose is a structural polymer of glucose. Glycogen and starch are storage homopolysaccharides.

(171) Answer : (4)

Solution:

Alkaloids – Morphine, codeine
Lectins – Concanavalin A
Essential oils – Lemon grass oil

(172) Answer : (3)

Hint:

Ion necessary for blood clotting

Solution:

Element	% Weight of Earth's crust	% Weight of human's body
Hydrogen (H)	0.14	9.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	Very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	Negligible

(173) Answer : (3)

Solution:

Inulin is a homopolysaccharide, that is found in the root of dahlia plant.

(174) Answer : (2)

Hint:

Component of sugars

Solution:

Element	% Weight of Earth's crust	% Weight of Human body
Hydrogen (H)	0.14	9.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	negligible

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(175) Answer : (2)

Hint:

Polymer of fruit sugar.

Solution:

Inulin is a polymer of fructose. Right end of polysaccharide is called the reducing end.

(176) Answer : (4)

Solution:

Cellulose is the most abundant structural polysaccharide in nature and is also a secondary metabolite categorised as a polymeric substance.

(177) Answer : (1)

Hint:

Each glycosidic bond is formed by formation of one water molecule.

Solution:

Both X and Y are hexose sugars which join to form a disaccharide (Z) by formation and release of one water molecule. So, reaction will be

$$C_6H_{12}O_6 + C_6H_{12}O_6 \xrightarrow{-H_2O} C_{12}H_{22}O_{11}$$

(178) Answer : (4)
Solution:

	Number of monomeric unit	Number of glycosidic bond(s) in them
Disaccharide	2	1
Pentasaccharide	5	4

(179) Answer : (3)
Hint:
Look for a lipid molecule amongst all
Solution:
Lipids are found in the retentate portion obtained upon the chemical analysis of a living tissue.
While adenylic acid, glucose and glycine are monomers which will be found in the filtrate portion.

(180) Answer : (4)
Hint:
Identify the molecule which is a monomer.
Solution:
Inulin is a polymer of fructose, while insulin is a polymer of amino acids. Lactose is a disaccharide. Polymers can be broken into monomers by addition of water (hydrolysis).

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